

MACKAR INLET, NU, CanadaWMO: **710800**Lat: **68.300N**Lon: **85.670W**Elev: **395**StdP: **96.67**Time Zone: **-6.00 (NAC)**Period: **82-92**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.7	-41.2	-41.0	0.1	-35.6	-39.9	0.1	-35.7	N/A	N/A	N/A	N/A	N/A	N/A	1.123

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.8	15.8	10.5	13.8	9.1	11.7	8.2	10.9	14.9	9.5	13.2	8.1	11.6	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.3	7.1	12.3	6.9	6.5	11.1	5.6	5.9	9.4	32.4	14.9	29.0	13.2	26.0	11.7	14.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	-45.0	17.9	1.9	2.0	-46.4	19.3	-47.5	20.5	-48.6	21.6	-50.0	23.0
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-14.9	-32.7	-32.3	-28.6	-20.4	-10.3	0.3	6.6	2.7	-2.5	-11.8	-22.1	-28.6		
	DBStd	14.80	5.78	6.02	6.01	6.00	5.12	4.19	4.50	3.75	4.25	5.74	6.48	7.22		
	HDD10.0	9103	1324	1186	1196	911	631	291	126	231	376	676	962	1196		
	HDD18.3	12118	1583	1419	1454	1161	889	540	362	485	626	934	1212	1454		
	CDD10.0	26	0	0	0	0	0	1	21	4	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	187	3	3	3	8	18	14	32	42	30	24	10	3		
	PrecMax	298	19	13	40	22	51	40	90	76	63	52	34	14		
	PrecMin	96	0	0	0	0	0	0	1	5	8	3	0	0		
	PrecStd	43	4	3	7	5	12	11	21	20	13	14	10	4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.9	-13.7	-10.8	-4.2	2.0	12.7	18.4	18.1	10.8	2.9	-4.0	-7.8		
		MCWB	-8.0	-14.2	-11.2	-4.6	1.2	8.9	11.5	13.6	8.0	2.9	-4.4	-8.2		
	2%	DB	-16.4	-17.2	-14.1	-7.6	0.3	10.8	16.6	13.9	8.3	-0.5	-5.7	-10.8		
		MCWB	-16.8	-17.4	-14.5	-8.0	-0.3	7.5	10.7	9.8	5.5	-0.7	-6.0	-11.1		
	5%	DB	-21.2	-20.6	-16.8	-10.1	-1.3	8.9	15.3	10.9	6.1	-2.5	-8.7	-14.6		
		MCWB	-21.6	-20.9	-17.2	-10.5	-1.7	6.3	10.0	8.2	4.5	-2.8	-9.1	-15.0		
	10%	DB	-25.1	-23.9	-19.9	-12.0	-2.9	6.5	13.6	8.2	4.0	-4.3	-12.5	-17.4		
		MCWB	-25.4	-24.1	-20.2	-12.5	-3.3	4.6	9.0	6.2	3.1	-4.6	-12.9	-17.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.9	-13.0	-11.0	-4.5	1.5	9.6	12.6	13.8	8.3	2.8	-4.3	-7.5		
		MCDB	-7.7	-12.8	-10.7	-4.3	1.9	12.1	16.9	17.8	10.9	3.0	-4.1	-7.3		
	2%	WB	-15.8	-16.6	-14.4	-7.8	-0.3	7.8	11.3	10.2	6.0	-0.5	-5.9	-10.8		
		MCDB	-15.5	-16.3	-14.2	-7.6	0.1	10.7	15.7	12.9	8.0	-0.4	-5.7	-10.3		
	5%	WB	-19.3	-19.6	-16.7	-10.5	-1.6	6.2	10.3	8.1	4.5	-2.7	-9.0	-14.4		
		MCDB	-19.0	-19.3	-16.4	-10.2	-1.2	8.6	14.6	10.7	5.8	-2.5	-8.5	-14.1		
	10%	WB	-22.6	-22.3	-19.7	-12.4	-3.3	4.7	9.2	6.1	3.1	-4.5	-12.9	-16.9		
		MCDB	-22.3	-22.0	-19.4	-12.0	-2.9	6.3	13.2	7.7	4.2	-4.3	-12.5	-16.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	4.8	5.0	5.0	3.8	3.5	4.8	4.1	3.1	4.0	4.5	4.4		
		MCDBR	8.4	6.2	5.8	6.0	4.4	6.5	6.3	7.8	4.2	3.8	6.5	5.5		
		MCWBR	7.9	6.1	5.7	6.0	4.5	4.7	3.8	5.4	3.2	3.8	6.5	5.4		
		MCDBR	8.0	7.1	5.6	5.9	4.4	5.9	6.4	7.2	4.2	3.8	6.5	5.6		
	5% WB	MCWBR	7.8	7.0	5.5	6.0	4.5	4.6	4.3	5.2	3.4	3.9	6.5	5.5		
(40) Clear-Sky Solar Irradiance	taub		0.153	0.198	0.241	0.305	0.316	0.301	0.335	0.309	0.269	0.222	0.147	N/A		
	taud		1.862	2.075	2.172	2.011	2.069	2.235	2.361	2.474	2.443	2.120	1.900	N/A		
	Ebn at Noon		120	633	810	826	865	896	850	841	788	567	113	0		
	Edh at Noon		7	44	77	123	131	115	99	79	61	41	6	0		
(44) All-Sky Solar Radiation	RadAvg		0.03	0.49	2.13	4.50	6.35	6.52	5.08	3.43	1.71	0.59	0.09	0.00		
	RadStd		0.00	0.03	0.07	0.14	0.28	0.35	0.48	0.27	0.21	0.08	0.01	0.00		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page